

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 41

AUTONOMOUS OMNIDIRECTIONAL "MULE" TUG MATRIX

The engines came out of the walls — the rails are dumb
Clamp the corner-castings — pivot 90° at any junction



Mule clamped to pod — yokes pivot 90° at the junction

PASSIVE RAILS — GRABBER CLAMPS — TORQUE POOLING

NO SWITCHES, NO TURNTABLES — THE GRID IS SOFTWARE

Basically just existing automated warehousing movement — his rider

GP-IM-041

Revision v1.0 — 21 August 2026

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VETTED: PASS — CONVENTIONAL ROBOTIC LOGISTICS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 41

PHASE 41: AUTONOMOUS OMNIDIRECTIONAL "MULE" TUG MATRIX — STATUS: CURRENT. The engines came out of the walls. The 500-metre Tri-Truss rail channels carry no linear motors, no rack-gears, no cables, no propulsion electronics — passive, dumb, indestructible structure that cannot be terminally damaged by a strike or an internal explosion. All the intelligence and torque lives in the Mules: high-torque robotic drive cubes scaled to the Phase 19 pod envelope, clamping onto structural corner-castings with symmetric grabber arms, gripping the rail lips with deep-flanged drive wheels on 90-degree rotational yokes, pooling torque through Phase 7 prism-laser coordination when one module needs a locomotive train. His own rider on the contents line: basically just normal existing automated warehousing movement.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-041, v1.0 — 21 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the Mule law as seated (contents line, the three design bodies, the origin — verbatim) — §2 why it works (passive-rail logic — graded) — §3 the system in the spine — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE: 'Phase 41: **Pod Movement-Autonomous Omnidirectional "Mule" Matrix.**

Eliminates all embedded rail propulsion — active linear motors, rack-gears, cables, and propulsion electronics — from the 500m structural hull rails, keeping the Tri-Truss spine channels fully passive and immune to terminal damage during structural strikes or internal explosions. Autonomous omnidirectional Mule locomotive pods provide all pod and container haulage, hot-swapping shuttle cargo in minutes (Phase 111) and recovering dropped parachutes (parachutes on earth) (Phase 116).'

THE CORE OBJECTIVE: 'Core Objective: Elimination of Embedded Rail Propulsion via Autonomous Mules'

PASSIVE INFRASTRUCTURE CORRIDOR RAILS (VERBATIM)

- Design Law: elimination of active linear motors, rack-gears, cables, or propulsion electronics inside the 500m structural hull rails.
- **Mechanical Resilience:** Keeps Tri-Truss spine channels entirely passive and immune to terminal damage during structural strikes or internal explosions.

AUTOMATED "MULE" LOCOMOTIVE PODS (VERBATIM)

- **Architecture:** High-torque, independent robotic drive cubes scaled to Phase 19 external dimensions to traverse all standard rail channels.
- **Clamping Mechanics:** Integrated high-strength mechanical grabber arms execute symmetric lock-on connections to the structural corner-castings of any habitation, cargo, or fuel pod.
- **Synchronized Train Link:** Multiple Mule units can execute multi-point clamping on one module, pooling torque vectors via Phase 7 prism-laser coordination as a single heavy-lift locomotive train.

OMNIDIRECTIONAL MOTOR-DRIVE WHEEL INTERFACE (VERBATIM)

- Rail Integration: Deep-flanged, high-traction drive wheels that clamp the upper guide lips of the Tri-Truss rails.
- **90-Degree Pivot Vectors:** Drive assemblies are mounted on independent rotational yokes. Allows the Mule to halt at 3D multi-deck junctions, rotate drive wheel geometry 90°, and execute cross-axis lateral translations without track switches or turntables.

ORIGIN OF THOUGHT — PHASE 41 (VERBATIM)

Archer, 3:11 pm, 16 July — asked whether pods would ride embedded linear motors or chain-drives: "Easier than that. Several pod-style connected forklifts have motor drive wheels as their rail lock system — omnidirectional, almost like an industrial package grabber in an automated warehouse."

Why it matters, in plain English: if the engines live in the walls, a single warped rail section freezes the whole deck. Archer moved the engines out of the walls entirely. The rails are dumb, passive, indestructible channels; all the intelligence and torque lives in swappable robot Mules that grab pods like warehouse forklifts grab pallets. A burnt-out Mule never traps a pod — a spare drives up the passive track and takes over. The warehouse industry already proved this model at Amazon scale; Archer scaled it to a Starship.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PASSIVE-RAIL INVERSION, AFFIRMED: embedded linear motors make every metre of rail a powered machine — one warped or burnt section freezes the deck. Passive channels carry nothing to fail; the failure domain shrinks from 500 metres of track to the Mules, which are swappable. This is why automated warehouses put the motors in the mover, not the floor.

THE CLAMP ARCHITECTURE, AFFIRMED: symmetric grabber arms onto structural corner-castings is the container-handling answer the ports of Earth already run — lock onto the strong points, never the skin. Multi-Mule torque pooling under coordinated control is conventional multi-agent robotics.

THE 90-DEGREE YOKE, AFFIRMED: rotating the drive assemblies at a junction kills the track switch and the turntable — the rail net becomes a grid, not a tree, and any path through it is a software decision.

THE DISTINCTION THAT KEEPS THE RECORD HONEST (the vet's, seated): the Mule replaces embedded rail HAULAGE, not the Mayernick magnetic-track system — the document separates those functions, and this chapter keeps them separate.

§3 — THE SYSTEM IN THE SPINE

- THE RAILS: entirely passive Tri-Truss spine channels — no active linear motors, rack-gears, cables or propulsion electronics inside the 500m hull rails; immune to terminal damage in a strike or internal explosion.
- THE MULES: high-torque independent robotic drive cubes scaled to Phase 19 external dimensions — they traverse every standard rail channel.
- THE CLAMPS: high-strength mechanical grabber arms executing symmetric lock-on to the structural corner-castings of any habitation, cargo or fuel pod.
- THE TRAIN: multiple Mules multi-point clamping one module, pooling torque vectors via Phase 7 prism-laser coordination as a single heavy-lift locomotive.
- THE WHEELS: deep-flanged high-traction drive wheels clamping the upper guide lips of the Tri-Truss rails — the under-clamp killed 21 August 2026 (§1): nothing connects under or through the rails.
- THE YOKES: independent rotational drive assemblies — halt at a 3D multi-deck junction, rotate 90 degrees, execute the cross-axis translation; no track switches, no turntables.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ENGINES-OUT-OF-THE-WALLS DOCTRINE (the origin, seated): 'if the engines live in the walls, a single warped rail section freezes the whole deck' — so the engines left the walls; the rails are dumb,

passive, indestructible channels and the intelligence is swappable (Archer, 16 July).

- THE SHIPPING-CONTAINER FRAME DOCTRINE (seated 21 August 2026, his ruling): pods are built like shipping containers — posts and rails proud of the walls, and the frames are where locking takes place, never the skin (the corner-casting clamp law stands on exactly this). When two frames meet, the gap created between the walls is how and why adjoining doors easily fit, seal and work. The pod corners are just posts in the room every 3 metres — a square room of 4 pods has a centre post that is 4 corner posts together — and there can NEVER be removal of posts for larger space in these areas: they form part of the ship frame. Only non-pod constructed areas — the bridges and the landing bay — do not have the posts.
- THE WAREHOUSE RIDER (his, on the contents line, verbatim): 'Basically just existing Automated warehousing movement.' — the novelty is the application, not the mechanism, and the phase says so itself.
- THE GRID-NOT-TREE DOCTRINE (seated in the law): no track switches, no turntables — the yoke pivot makes every junction a software decision.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Certify the channels: Tri-Truss rail channels verified passive — no powered hardware seated anywhere in the 500m runs; lip gauge and straightness mapped.
- STEP 2 — Build the Mule: the drive cube to the Phase 19 envelope; deep-flanged wheels against the rail lips; traction and braking benched at rated pod mass.
- STEP 3 — Prove the clamp: grabber-arm lock-on to the corner-castings; symmetric load test to the heaviest fuel pod; failure mode must be grip-held, never release.
- STEP 4 — Pivot the yokes: 90-degree junction rotations under load; the cross-axis translation walked before it is run.
- STEP 5 — Pool the train: multi-Mule clamping and Phase 7 prism-laser torque coordination — two units, then four, then the heavy-lift consist; the minutes-scale shuttle cargo hot-swap (Phase 111) and parachute recovery (Phase 116) drills run to routine.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 13 — the Tri-Truss spine (GP-IM-013): the passive rail channels and the guide lips the wheels grip.

- PHASE 19 — the standardized pods (GP-IM-019): the external envelope the Mule is scaled to; the corner-castings it clamps.
- PHASE 7 — the prism-laser coordination (GP-IM-007): the multi-Mule torque-pooling link.
- PHASE 111 — the shuttle cargo hot-swap: minutes-scale turnaround, seated in the contents line.
- PHASE 116 — the parachute recovery: dropped chutes on Earth, the Mules' ground job.
- PHASE 46 — the rescue loop (GP-IM-046): the rescue pod's autonomous circularization rides integrated Mule drive wheels.

§8 — DO-NOT-BUILD

- NEVER embed propulsion in the rails — no linear motors, rack-gears, cables or track electronics; the passive-channel law is the phase (§1).
- NEVER clamp a pod by its skin — corner-castings only; the skin is pressure vessel, not structure (§1).
- NEVER connect under or through the rails — the rail slots are full of the silicone sausages, and under would require human intervention; the Mule is autonomous (his ruling, 21 August 2026).
- NO track switches or turntables — the yoke pivot is the junction (§1).
- NEVER conflate the Mule with the Mayernick track — haulage and the magnetic-track system are separate functions, kept separate (§5).

§9 — OWED

THE BENCH, OWED: the vet found no physics owed — the engineering register stands: corner-casting clamp load rating at the heaviest pod; upper-lip wheel traction and braking curves at rated mass; multi-Mule torque-pooling authority limits; yoke pivot cycle life under load. And one spec landed on the register 21 August 2026: THE WHEEL-RAIL INTERFACE GEOMETRY — the under-clamp is killed ('you cannot connect under or through the rails'); the final positive interface is his spec, owed. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-041, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 21 August 2026, fourth of the 'next 10' shift. First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i

will read them when i get back' — the order that opened the chapter program; the follow-on order (seated in sitting 537): 'ok next 10'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 41